

EXP NO:10

STUDENT APP USING SQLITE

AIM:

To create an Student Application using Android studio for maintaining student details in sqlite.

ALGORITHM:

Step 1: Start the Android Studio.

Step 2: File -> New -> New Project ->Application Name ->StudentAppNext->Finish.

Step 3: Design the App (activity_main.xml) by adding the following controls.
4Button,2 Edit Text, 2 Text View.

Step 4: Assign the name for onClick event as inf,delf,updatef,viewf.Step 5:Write coding for MainActivity.java.

Step 5: Run the program Build -> BuildBundles/apk(s)->locate.

Step 6: Now the App is installed and verified in Bluestack.

Step 7:Close the project.

CODE: XML

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<RelativeLayout
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
    xmlns:tools="http://schemas.android.com/tools"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    tools:context=".MainActivity">
```

```
<TextView
```

```
    android:id="@+id/textView
```

```
    "
```

```
    android:layout_width="84dp"
```

```
    "
```

```
    android:layout_height="51d
```

```
    p"
```

```
    android:layout_alignParentStart="true"
```

```
    android:layout_alignParentLeft="true"
```

```
    android:layout_alignParentTop="true"
```

```
    android:layout_marginStart="19dp"
```

```
    android:layout_marginLeft="19dp"
```

```
    android:layout_marginTop="10dp"
```

```
    android:text="name"
```

```
<TextView
```

```
    android:id="@+id/textView
```

```
    2"
```

```
    android:layout_width="75dp"
```

```
    "
```

```
    android:layout_height="44d
```

```
    p"
```

```
    android:layout_alignEnd="@+id/textVie
```

```
    w" android:layout_alignParentTop="true"
```

```
    android:layout_marginTop="92dp"
```

```
    android:layout_marginEnd="3dp"
```

```
    android:text="rno" />
```

```
<EditText
    android:id="@+id/editText"

    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignBottom="@+id/text
View"
    android:layout_marginStart="31dp"
    android:layout_marginBottom="-3dp"
    android:layout_toEndOf="@+id/textVie
w2" android:ems="10"
    android:inputType="textPersonName"
/>
```

```
<EditText
    android:id="@+id/editText2
"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignEnd="@+id/editText"
    android:layout_alignBottom="@+id/textView2"
    android:ems="10"
    android:inputType="textPersonName"
/>
```

```
<Button
    android:id="@+id/button"

    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    "

    android:layout_alignParentStart="true"
    "

    android:layout_alignParentLeft="true"
    android:layout_alignParentTop="true"
    android:layout_marginStart="51dp"
    android:layout_marginLeft="51dp"
    android:layout_marginTop="181dp"
    android:onClick="inf"
    android:text="insertion" />
```

```
<Button
    android:id="@+id/button2"

    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    "

    android:layout_alignTop="@+id/button"
    "

    android:layout_alignParentEnd="true"
    android:layout_alignParentRight="true"
    " android:layout_marginTop="3dp"
    android:layout_marginEnd="64dp"
    android:layout_marginRight="64dp"
    android:onClick="delf"
    android:text="deletion" />
```

```
<Button
    android:id="@+id/button3"
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignEnd="@+id/button"
    android:layout_alignParentBottom="true"
    android:layout_marginEnd="-8dp"
    android:layout_marginBottom="118dp" android:onClick="updatef"
    android:text="updation" />
```

```
<Button
    android:id="@+id/button4"
```

```

        android:layout_width="wrap_content"
        "
        android:layout_height="wrap_content"
        "
        android:layout_alignEnd="@+id/text
        View"
        android:layout_alignParentTop="true"
        "
        android:layout_marginTop="346dp"
        android:layout_marginEnd="-250dp"
        android:onClick="viewf"

    android:text="viewing" />

</RelativeLayout>

```

CODE: MainActivity.java

```

package com.example.admin.studentapp;

import android.content.Context;
import android.database.Cursor;

import
android.database.sqlite.SQLiteDatabase;
import
android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import
android.widget.Button;
import
android.widget.EditText;
import
android.widget.TextView;
import
android.widget.Toast;

public class MainActivity extends
    AppCompatActivity { EditText name,rno;
    Button i,d,u,v;
    SQLiteDatabase
    db;

```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    name=(EditText)findViewById(R.id.edit
Text); rno=(EditText)
    findViewById(R.id.editText2);
    i=(Button)findViewById(R.id.button);
    d=(Button)findViewById(R.id.button2);
    u=(Button)findViewById(R.id.button3);
    v=(Button)findViewById(R.id.button4);
```

```

        db=openOrCreateDatabase("StudentDB", Context.MODE_PRIVATE, null);
        db.execSQL("CREATE TABLE IF NOT EXISTS student(name
        VARCHAR,rno
        VARCHAR);");
    }

    public void inf(View v)
    {
        db.execSQL("INSERT INTO student
        VALUES('"+name.getText()+"','"+rno.getText()+"');");

        Toast.makeText(getApplicationContext(),"insertion
        success",Toast.LENGTH_LONG).show();
        clearText();
    }

    public void delf(View v)
    {
        Cursor c=db.rawQuery("SELECT * FROM student WHERE
        rno='"+rno.getText()+"'", null);
        if(c.moveToFirst())
        {
            db.execSQL("DELETE FROM student WHERE
            rno='"+rno.getText()+"'");

            Toast.makeText(getApplicationContext(),"record
            deleted",Toast.LENGTH_LONG).show();
        }
        else
        {
            Toast.makeText(getApplicationContext(),"inv
            alid roll number",Toast.LENGTH_LONG).show();
        }
        clearText();
    }

    public void viewf(View v)
    {

```

```
Cursor c=db.rawQuery("SELECT * FROM student WHERE  
rno='"+rno.getText()+"'", null);  
if(c.moveToFirst())  
{  
    name.setText(c.getString(0));
```

```

    }
    else
    {
        Toast.makeText(getApplicationContext(),"No such
record",Toast.LENGTH_LONG).show();
        clearText();
    }
    public void updatef(View v)
    {

        Cursor c=db.rawQuery("SELECT * FROM student WHERE
rno='"+rno.getText()+"'", null);

        if(c.moveToFirst()) {
            db.execSQL("UPDATE student SET name='" + name.getText() + "'
WHERE rno='" + rno.getText() + "'");

            Toast.makeText(getApplicationContext(),"Updation
success",Toast.LENGTH_LONG).show();

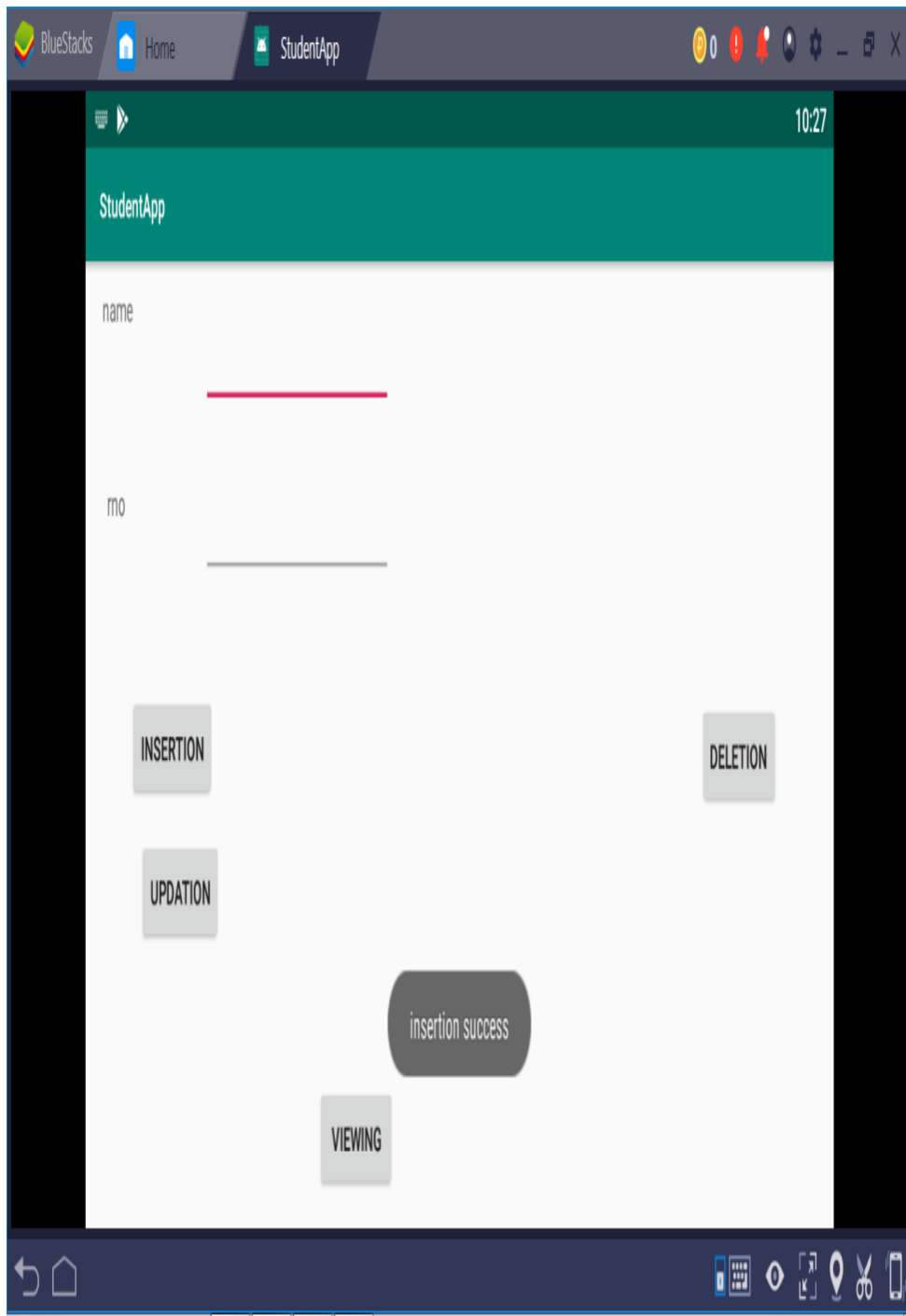
        }
        else {
            Toast.makeText(getApplicationContext(),"No such
record",Toast.LENGTH_LONG).show();
        }
        clearText();
    }

    public void clearText()
    {
        rno.setText("");
        name.setText("");
        name.requestFocus();
    }
}

}

```

OUTPUT:



RESULT:

Thus, The Android Application has been developed and run successfully